

Assignment

High-Level Data Warehouse Design from an OLTP Movie Rental System

Course: Data Warehousing / Data Architecture

Assignment Type	Individual or Group
Main Topic	From OLTP Schema to Dimensional Model and ETL Design

OLTP schema link: [Open the OLTP movie rental schema](https://drive.google.com/file/d/1UxR65upm350BBJkEODJQ3h2LQKGrw5-L/view?usp=drive_link)

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1. Assignment Context

You are given an OLTP database schema for a movie rental business. The schema supports daily operational activities such as registering customers, managing films, storing inventory, recording rentals, processing payments, managing staff, and maintaining store and location information. Although this OLTP database is suitable for transaction processing, it is not optimized for analytical reporting. Business managers need a data warehouse that can help them analyze rental activity, revenue, customer behavior, film popularity, store performance, and trends over time.

2. Assignment Objective

In this assignment, you will act as a data warehouse designer. Your task is to study the OLTP schema and propose a high-level data warehouse design using dimensional modeling principles. You are not required to implement the full data warehouse physically. The focus is on analysis, design decisions, dimensional modeling, and ETL planning.

- Understand the difference between OLTP and data warehouse design.
- Identify analytical business questions from an operational database.
- Determine suitable fact tables and dimension tables.
- Define the grain of fact tables.
- Design a high-level dimensional model.
- Explain how ETL can extract, transform, and load data from the OLTP system into the data warehouse.
- Justify your design decisions clearly.

3. Assignment Requirements

Your submission should answer the following three main questions. The aim is not to copy the OLTP schema, but to reorganize it into an analytical structure suitable for reporting and decision-making.

Question 1: What Business Questions Should the Data Warehouse Answer?

Before designing the data warehouse, identify the type of analysis the business needs. Study the OLTP schema and propose a set of analytical questions that the data warehouse should support. Your questions may focus on rental activity, revenue analysis, film popularity, customer behavior, store performance, staff performance, location-based analysis, and time-based trends.

Examples of possible business questions include:

1. Which films are rented most frequently?
2. Which films generate the highest revenue?
3. Which film categories are most popular?
4. Which stores generate the highest number of rentals?
5. Which stores generate the highest revenue?
6. Which customers rent the most films?
7. Which customers generate the highest revenue?
8. How does rental activity change over time?
9. How does revenue change by month, quarter, or year?
10. Which staff members process the highest number of rentals or payments?
11. Which cities or countries generate the highest customer activity?
12. What is the average rental duration for different films or categories?
13. Which films are returned late most often?
14. How does store performance differ by location?
15. What are the most active customer locations?

You are expected to select a reasonable set of business questions and briefly explain why these questions are important for decision-making.

Question 2: What Fact Tables and Dimension Tables Should Be Designed?

After identifying the business questions, propose a dimensional model that can answer them. Identify the main business processes in the OLTP schema and decide which processes should become fact tables.

Your design should discuss:

- What are the main business processes?
- What fact tables are needed?
- What is the grain of each fact table?
- What measures should be stored in each fact table?
- What dimension tables are needed?
- Which dimensions are shared between fact tables?
- Should the design be a star schema, snowflake schema, or hybrid schema?
- Which OLTP tables are used to build each dimension?

At a high level, possible fact tables may be related to:

- Film rental transactions
- Payment transactions

- Inventory or film availability
- Customer rental activity

Possible dimension tables may include:

- Date
- Customer
- Film
- Category
- Store
- Staff
- Location
- Language
- Actor

You are not expected to copy all OLTP tables into the data warehouse. A good dimensional model should simplify the operational schema and organize it around analysis.

Your final dimensional model should clearly show fact tables, dimension tables, primary keys, foreign keys, main measures, and relationships between facts and dimensions.

Question 3: How Should ETL Fill the Data Warehouse?

The final part of the assignment is to propose a high-level ETL process. Describe how data will be moved from the OLTP system into the data warehouse.

3.1 Extract

Identify which OLTP tables should be extracted from the source database. Examples of source tables may include:

- rental
- payment
- customer
- film
- inventory
- store
- staff
- address
- city
- country
- category
- film_category
- language
- actor
- film_actor

You should explain why each selected source table is needed.

3.2 Transform

Explain what transformations are required before loading the data warehouse. Your discussion may include:

- Joining related OLTP tables to create meaningful dimensions.
- Combining first name and last name into full names.
- Cleaning missing, duplicated, or invalid data.
- Converting operational IDs into warehouse surrogate keys.
- Creating date keys from rental dates, return dates, and payment dates.
- Calculating measures such as rental duration, rental count, payment amount, and revenue.
- Standardizing text values such as city, country, language, and status.
- Handling many-to-many relationships such as film-category and film-actor.
- Handling missing return dates for films not yet returned.
- Detecting late returns by comparing actual rental duration with expected rental duration.
- Preparing for historical changes in customer, staff, store, or film attributes.

You do not need to write full ETL code, but your explanation should be clear enough to show how the data warehouse would be populated.

3.3 Load

Explain how transformed data should be loaded into the data warehouse. Your answer should discuss:

- Why dimension tables are usually loaded before fact tables.
- How surrogate keys are generated for dimensions.
- How fact tables reference dimension surrogate keys.
- How new records are inserted.
- How changed records may be handled.
- How data quality checks are applied before or after loading.

A reasonable loading order may be: date dimension, customer dimension, film-related dimensions, store and staff dimensions, rental fact table, and payment fact table.

4. Expected Deliverables

Submit a report containing the following sections:

- **Section 1: Introduction:** Briefly describe the purpose of the assignment and the difference between the OLTP system and the proposed data warehouse.
- **Section 2: Business Questions:** List the analytical business questions that your data warehouse should answer. Explain their importance.
- **Section 3: Dimensional Model Design:** Describe your proposed fact tables and dimension tables. For each fact table, include the business process, grain, measures, and related dimensions. For each dimension table, include the dimension name, main attributes, source OLTP tables, and purpose of the dimension.
- **Section 4: Dimensional Model Diagram:** Provide a clear diagram showing your proposed star, snowflake, or hybrid schema. The diagram should clearly show the relationship between fact tables and dimension tables.

- **Section 5: ETL Design:** Describe the Extract, Transform, and Load process used to populate the data warehouse.
- **Section 6: Data Quality Considerations:** Define a small set of data quality rules that should be checked during ETL.
- **Section 7: Conclusion:** Summarize your proposed solution and explain how it supports business decision-making.

5. Suggested Report Structure

1. Introduction
2. Business Questions
3. Dimensional Model Design
 - 3.1 Selected Business Processes
 - 3.2 Fact Tables
 - 3.3 Dimension Tables
4. Dimensional Model Diagram
5. ETL Design
 - 5.1 Extract
 - 5.2 Transform
 - 5.3 Load
6. Data Quality Considerations
7. Conclusion

6. Important Notes

You should submit the answers, codes, documentations on GitHub.

Just share the link of the GitHub.